

Questions with Answer Keys

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Q1 - Single Correct

Let x and y be the positive real numbers such that $\log_x y = -\frac{1}{4}$, then value of the expression $\log_x (xy^5) - \log_y \left(\frac{x^2}{\sqrt{y}}\right)$, is

- (1) $\frac{31}{4}$
- (2) $\frac{33}{4}$
- (3) $\frac{35}{4}$
- (4) 8

Q2 - Single Correct

If the equation $|x^2 + 4x + 3| + 2m - mx = 0$ has exactly three solutions, then the value of $(m + 8)$ can be

- (1) $\sqrt{15}$
- (2) $-\sqrt{15}$
- (3) $\sqrt{40}$
- (4) $-\sqrt{60}$

Q3 - Single Correct

If $(x^2 - 5x + 4)(y^2 + y + 1) < 2y$ for all real y , then x belongs to the interval $(2, b)$, then b can be

- (1) 3
- (2) 4
- (3) 5
- (4) 6

Q4 - Single Correct

If $x^4 + 3 \cos(ax^2 + bx + c) = 2(x^2 - 2)$ has two solutions with $a, b, c \in (2, 5)$, then the maximum value of $\frac{ac}{b^2}$ is

- (1) 1
- (2) 2
- (3) 3
- (4) $\frac{16}{3}$

Q5 - Single Correct

Let $p(x) = x^2 + \frac{4x}{3} + \log_{10}(4.9)$, $A = \prod_{i=1}^{12} p(a_i)$, where $a_1, a_2, \dots, a_{12}$ are positive reals and $B = \prod_{j=1}^{13} p(b_j)$, where $b_1, b_2, \dots, b_{13}$ are non-positive reals, then which one of the following is always correct?

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(1) $A > 0$ and $B > 0$

(2) $A > 0$ and $B < 0$

(3) $A < 0$ and $B > 0$

(4) $A < 0$ and $B < 0$

Q6 - Single Correct

If the equation $2^x + 2^{-x} = 2k$ has exactly one real solution, then sum of all integral values of k in $[-100, 100]$ is equal to

(1) 5050

(2) 10100

(3) 0

(4) -5050

Q7 - Single Correct

Let x be a real number which satisfy $\log_3 x = 1 + \frac{1}{2} \sin \theta$, where $\theta \in [0, 2\pi]$, then $\left|x - \frac{3}{2}\right| + \left|\frac{11}{2} - x\right|$ is equal to

(1) 7

(2) $2x - 7$

(3) $7 - 2x$

(4) 4

Q8 - Single Correct

Let $A = \{x \mid x^2 + (m-1)x - 2(m+1) = 0, x \in R\}$ and $B = \{x \mid x^2(m-1) + mx + 1 = 0, x \in R\}$. Number of values of m such that $A \cup B$ has exactly 3 distinct elements, is

(1) 4

(2) 5

(3) 6

(4) 7

Q9 - Single Correct

The inequalities $y(-1) \geq -4$, $y(1) \leq 0$ and $y(3) \geq 5$ are known to hold for $y = ax^2 + bx + c$ then the least value of a is

(1) $-\frac{1}{4}$

(2) $-\frac{1}{3}$

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(3) $\frac{1}{4}$

(4) $\frac{1}{8}$

Q10 - Single Correct

The number of integral solutions of equation $y^2 (5x^2 + 1) = 25 (2x^2 + 13)$, where $x, y \in I$, are

(1) 2

(2) 4

(3) 8

(4) Infinite

Q11 - Multiple Correct

If the equation $x^2 + 2|a|x + 4 = 0$ has integral roots, then the minimum value of a is

(1) $-\frac{5}{2}$

(2) -1

(3) 0

(4) 1

Q12 - Multiple Correct

If a, b are positive real numbers such that $a - b = 2$, then the smallest value of the constant L for which

$\sqrt{x^2 + ax} - \sqrt{x^2 + bx} < L$ for all $x > 0$, is

(1) $\frac{1}{2}$

(2) $\frac{1}{\sqrt{2}}$

(3) 1

(4) 2

Q13 - Multiple Correct

Let a, b and c be three distinct non-zero real numbers satisfying the system of equation

$\frac{1}{a} + \frac{1}{a-1} + \frac{1}{a-2} = 1, \frac{1}{b} + \frac{1}{b-1} + \frac{1}{b-2} = 1, \frac{1}{c} + \frac{1}{c-1} + \frac{1}{c-2} = 1$, then

(1) $a + b + c = 6$

(2) $abc = 2$

(3) $(1-a)(1-b)(1-c) = 1$

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(4) $(a-2)(b-2)(c-2) = 2$

Q14 - Multiple Correct

The roots of equation $x^5 - 40x^4 + \alpha x^3 + \beta x^2 + \gamma x + \delta = 0$ are real and in GP. If the sum of their reciprocals is 10, then δ can be

(1) -32

(2) $-\frac{1}{32}$

(3) 32

(4) $\frac{1}{32}$

Q15 - Multiple Correct

Let $f(x) = (x^2 + 2x + 3)^2 + 2(x^2 + 2x + 3) + 3$, then which of the following statement(s) is/are correct?

(1) The equation $f(x) = 0$ has no real roots

(2) The equation $f(x) = 0$ has two real and two imaginary roots

(3) The minimum value of $f(x)$ is 11

(4) The minimum value of $f(x)$ is 12

Q16 - Multiple Correct

The equation $x^{\frac{3}{4}(\log_2 x)^2 + \log_2 x - \frac{5}{4}} = \sqrt{2}$ has

(1) atleast one real solution

(2) exactly three solutions

(3) exactly one irrational solution

(4) complex roots

Q17 - Multiple Correct

If α, β are the roots of $x^2 + px + q = 0$ and α^4, β^4 are the roots of $x^2 - rx + s = 0$ then the equation

$x^2 - 4qx + 2q^2 - r = 0$ has always

(1) two real roots

(2) two positive roots

(3) two negative roots

(4) one positive and one negative roots

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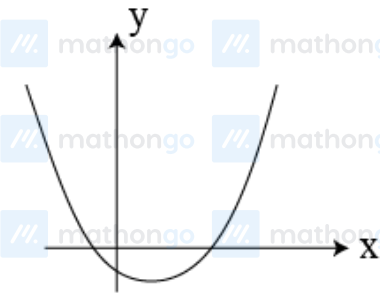
Q18 - Multiple Correct

If $a, b \in R$ and $ax^2 + bx + 6 = 0, a \neq 0$ does not have two distinct real roots, then

- (1) Minimum possible value of $3a + b$ is -2
- (2) Minimum possible value of $3a + b$ is 2
- (3) Minimum possible value of $6a + b$ is -1
- (4) Minimum possible value of $6a + b$ is 1

Q19 - Multiple Correct

The graph of the quadratic polynomial $y = ax^2 + bx + c$ is as shown in the figure . Then



- (1) $b^2 - 4ac > 0$
- (2) $b < 0$
- (3) $a > 0$
- (4) $c < 0$

Q20 - Multiple Correct

If the quadratic equation $(ab - bc)x^2 + (bc - ca)x + ca - ab = 0, a, b, c \in R$, has both the roots equal, then

- (1) both roots are equal to 0
- (2) both roots are equal to 1
- (3) a, c, b are in harmonic progression
- (4) $ab^2c^2, b^2a^2c, a^2c^2b$ are in arithmetic progression

Q21 - Multiple Correct

One real solution of the equation $(x^2 + 2)^2 + 8x^2 = 6x(x^2 + 2)$ is

- (1) $2 + \sqrt{2}$
- (2) $2 - \sqrt{2}$

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(3) $3 + \sqrt{3}$

(4) $3 - \sqrt{3}$

Q22 - Multiple Correct

If the roots of the equation $x^3 + bx^2 + cx - 1 = 0$ form an increasing G.P., then

(1) $b + c = 0$

(2) $b \in (-\infty, -3)$

(3) one of the roots is 1

(4) one root is smaller than 1 and one root is more than 1.

Q23 - Multiple Correct

$$\frac{\pi^e}{x-e} + \frac{e^\pi}{x-\pi} + \frac{\pi^{\pi+e}}{x-\pi-e} = 0$$
 has

(1) one real root in (e, π) and other in $(\pi - e, e)$ (2) one real root in (e, π) and other in $(\pi, \pi + e)$ (3) two real roots in $(\pi - e, \pi + e)$

(4) No real roots

Q24 - Multiple Correct

The value of the expression $x^4 - 8x^3 + 18x^2 - 8x + 2$ when $x = 2 + \sqrt{3}$ is

(1) 0

(2) 1

(3) 2

(4) 3

Q25 - Multiple Correct

If α, β are the roots of the equation $ax^2 + bx + c = 0$, then the equation $ax^2 - bx(x-1) + c(x-1)^2 = 0$ has roots

(1) $\frac{\alpha}{1-\alpha}, \frac{\beta}{1-\beta}$

(2) $\alpha - 1, \beta - 1$

(3) $\frac{\alpha}{\alpha+1}, \frac{\beta}{\beta+1}$

(4) $\frac{1-\alpha}{\alpha}, \frac{1-\beta}{\beta}$

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Q26 - Comprehension Type

Passage I (For Question 26, 27)

Let $p(x) = x^5 - 9x^4 + px^3 - 27x^2 + qx + r$ (where, $p, q, r \in R$) be divisible by x^2 and α, β are the positive roots of the equation, $\frac{p(x)}{x^2} = 0$

The value of $(p + q + r)$ is

- (1) 9
- (2) 27
- (3) 1
- (4) 108

Q27 - Comprehension Type

If $(\alpha - 1)$, $(\beta + 3)$ and $(\gamma + 7)$ are the first three terms of a sequence whose sum of first n terms is given by S_n , then

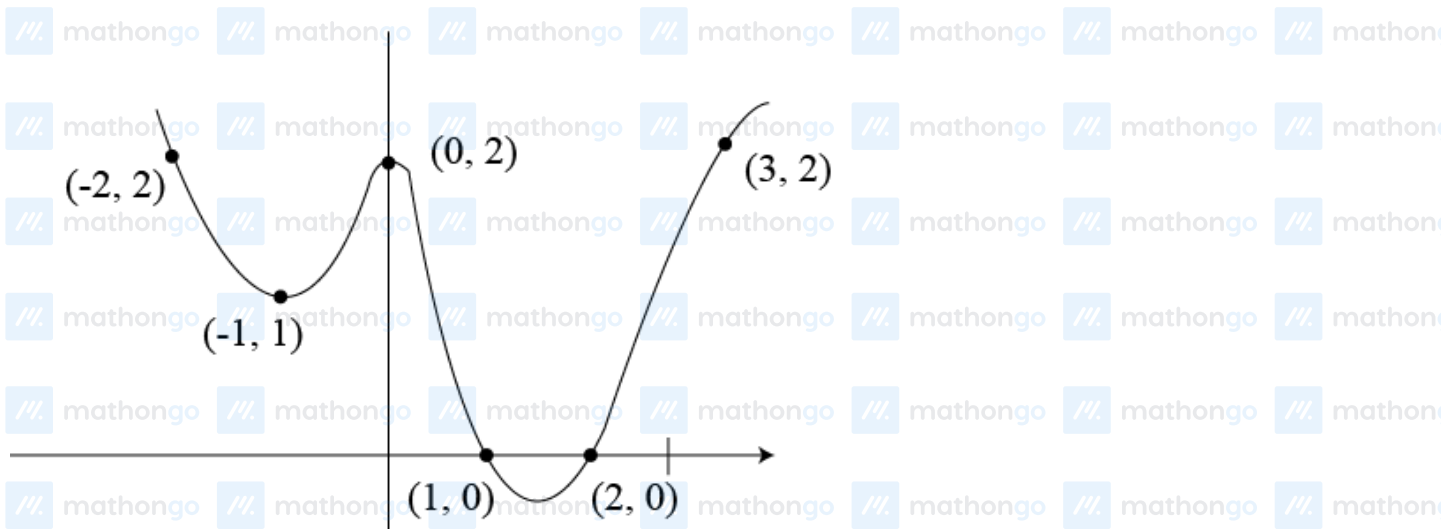
$\sum_{n=2}^{\infty} \frac{1}{\sqrt{S_n \cdot S_{n-1}}}$ is equal to

- (1) 1
- (2) $\frac{1}{4}$
- (3) $\frac{1}{2}$
- (4) 2

Q28 - Comprehension Type

Passage II (For Question 28, 29)

In the given figure, graph of $y = p(x) = x^4 + ax^3 + bx^2 + cx + d$ is given



The product of all imaginary roots of $p(x) = 0$, is

- (1) -2
- (2) -1
- (3) $-\frac{1}{2}$
- (4) None of these

Q29 - Comprehension Type

If $p(x) + c = 0$ has 4 distinct real roots α, β, γ and δ , then $[\alpha] + [\beta] + [\gamma] + [\delta]$ is equal to

- (1) -1
- (2) -2
- (3) 0
- (4) 1

Q30 - Comprehension Type

Passage III (For Question 30, 31)

Consider two quadratic trinomials and

$$f(x) = x^2 - 2ax + (a^2 - 1)$$

$$g(x) = (4b - b^2 - 5)x^2 - (2b - 1)x + 3b$$

where $a, b \in R$.

The values of a for which both roots of the equation $f(x) = 0$ are greater than -2 but less than 4, lie in the interval

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(1) $-1 < a < 3$

(2) $5 < a < \infty$

(3) $-2 < a < 0$

(4) None of the above

Q31 - Comprehension Type

If roots of the quadratic equation $g(x) = 0$ lie on either side of unity, then number of integral values of b is

(1) 1

(2) 2

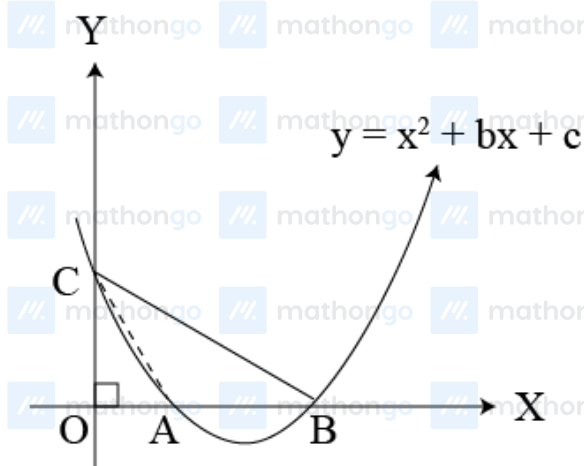
(3) 3

(4) 4

Q32 - Comprehension Type

Passage IV (For Question 32, 33)

In the given figure, $\triangle OBC$ is a right angled isosceles triangle in which AC is a median. Then, answer the following questions.



The roots of $y = ax^2 + bx + c$ are

(1) $\{1, 2\}$

(2) $\{2, 4\}$

(3) $\{\frac{1}{2}, 1\}$

(4) $\{4, 8\}$

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Q33 - Comprehension Type

The equation whose roots are $(\alpha + \beta)$ and $(\alpha - \beta)$, where $\alpha, \beta (\alpha > \beta)$ are roots obtained in previous question, is

(1) $x^2 - 4x + 3 = 0$

(2) $x^2 - 8x + 12 = 0$

(3) $4x^2 - 8x + 3 = 0$

(4) $x^2 - 16x + 48 = 0$

Q34 - Comprehension Type

Passage V (For Question 34, 35)

For $a, b \in \mathbb{R} - \{0\}$, $f(x) = ax^2 + bx + a$ satisfies $f\left(x + \frac{7}{4}\right) = f\left(\frac{7}{4} - x\right)$, $\forall x \in \mathbb{R}$. Also, the equation $f(x) = 7x + a$ has only one real and distinct solution.

The value of $(a + b)$ is

(1) 4

(2) 5

(3) 6

(4) 7

Q35 - Comprehension Type

The minimum value of $f(x)$ in $\left[0, \frac{3}{4}\right]$ is

(1) $-\frac{33}{8}$

(2) 0

(3) 4

(4) -2

Q36 - Matrix Match (Only one option is correct)

	Column I		Column II
i.	If $ax^2 - ax + 1 > 0$ for atleast one x , then the range of a is	p.	$[0, 4)$
ii.	If $x^3 - 3x + \frac{a}{2} = 0$ has three real and distinct roots, then the interval of $ a $ is	q.	$[0, 3]$
iii.	If $x^3 + ax^2 + ax + 1 = 0$ is an increasing function. then the interval of a may belong to	r.	$\mathbb{R}$
iv.	If quadratic equation $x^2 - 3ax + (a^2 - 9) = 0$ has roots of opposite sign, then a belongs to	s.	$(-3, 3)$

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(1)

i	ii	iii	iv
r	p	q	s

(2)

i	ii	iii	iv
r	p	s	q

(3)

i	ii	iii	iv
r	q	p	s

(4)

i	ii	iii	iv
s	p	q	r

Q37 - Matrix Match (Only one option is correct)

$f(x) = x^3 + ax^2 + bx + c$ has three distinct integer roots and

$(x^2 + 2x + 2)^3 + a(x^2 + 2x + 2)^2 + b(x^2 + 2x + 2) + c = 0$ has no real roots, then

	Column I		Column II
i.	$a =$	p.	0
ii.	$b =$	q.	2
iii.	$c =$	r.	3
iv.	If the roots of $f'(x) = k$ are equal, then $k =$	s.	-1

(1)

i	ii	iii	iv
r	q	p	s

(2)

i	ii	iii	iv
r	q	s	p

(3)

i	ii	iii	iv
s	q	s	r

(4)

i	ii	iii	iv
r	p	q	s

Q38 - Matrix Match (Only one option is correct)

The expression $y = ax^2 + bx + c$ (where $a, b, c \in R, a \neq 0$) represents a parabola, which cuts the X -axis at the points,

which are roots of the equation $ax^2 + bx + c = 0$

	Column I		Column II
i.	For $a = 1$ $c = 4$ if both the roots are greater than 2 then $b \in$	p.	4
ii.	For $a = -1$ $b = 5$ if the roots lie on either side of -1 then $c \in$	q.	8
iii.	For $b = 6$ $c = 1$ if one root is less than -1 and the other root greater than $-\frac{1}{2}$ then $a \in$	r.	10
		s.	No real value

(1)

i	ii	iii
s	q	p

Questions with Answer Keys

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(2)

i	ii	iii
s	q r	s

(3)

i	ii	iii
p s	q s	p

(4)

i	ii	iii
r s	q s	p s

Q39 - Integer Type

If $x^2 - |a - 1|x + 1 = |x|$ has exactly three real roots, then sum of the values of a is

Q40 - Integer Type

If $x^5 - x^3 + x = a$, then $x^6 \geq 2a - k$, when value of k is equal to

Q41 - Integer Type

Find the sum of all real values of k for which the equation $|x^2 - (7 + k^2)x + 7k^2| + \sqrt{(x - 3)(x - 3k + 2)} = 0$ has atleast one real solution

Q42 - Integer Type

If the system of equation $3x^2 + 2x - 1 < 0$ and $(3a - 2)x - a^2x + 2 < 0$ possesses solution, find the least natural number a .

Q43 - Integer Type

If $f(x) = x^3 - 3x + 1$, then find the number of distinct real roots of the equation $f[f(x)] = 0$.

Q44 - Integer Type

Let $f(x) = x^4 + ax^3 + bx^2 + cx + d$ be a polynomial with real coefficients and real roots $|f(i)| = 1$, then find the value of $(a + b + c + d)$.

Q45 - Integer Type

The smallest value of k , for which both the roots of the equation $x^2 - 8kx + 16(k^2 - k + 1) = 0$ are real, distinct and have values atleast 4, is

Q46 - Integer Type

Questions with Answer Keys

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Find the number of digits in the sum of all integral values of a in $[1, 100]$ for which the equation

$$x^2 - (a - 5)x + \left(a - \frac{15}{4}\right) = 0 \text{ has atleast one root greater than zero.}$$

Q47 - Integer Type

Let x_1 and x_2 be the real root of the equation $x^2 - kx + (k^2 + 7k + 15) = 0$. If the maximum value of $(x_1^2 + x_2^2)$ is $\frac{18}{x}$, then find the value of x .

Q48 - Integer Type

If all the solutions of the inequality $x^2 - 6ax + 5a^2 \leq 0$ are also the solutions of inequality $x^2 - 14x + 40 \leq 0$, then find the number of possible integral values of a .

Q49 - Integer Type

Find the number of integral values of a , so that the inequation $x^2 - 2(a + 1)x + 3(a - 3)(a + 1) < 0$ is satisfied by atleast one $x \in \mathbb{R}^+$.

Q50 - Integer Type

Let $f(x) = kx^2 + x(3 - 4k) - 12$. If the set of values of k for which $f(x) < 0, \forall x \in (-3, 3)$ and $f(-4) > 0$ is $(p, q]$, then find the value of $4p + 3q$.

Answer Key**Q1 (2)****Q2 (4)****Q3 (1)****Q4 (1)****Q5 (1)****Q6 (1)****Q7 (4)****Q8 (4)****Q9 (4)****Q10 (2)****Q11 (1)****Q12 (3)****Q13 (1,2,3,4)****Q14 (1,3)****Q15 (1,3)****Q16 (1,2,3)****Q17 (1,4)****Q18 (1,3)****Q19 (1,2,3,4)****Q20 (2,3,4)****Q21 (1,2)****Q22 (1,2,3,4)****Q23 (2,3)****Q24 (2)****Q25 (3)****Q26 (2)****Q27 (3)****Q28 (4)****Q29 (1)****Q30 (1)****Q31 (2)****Q32 (1)****Q33 (1)****Q34 (2)****Q35 (4)****Q36 (1)****Q37 (1)****Q38 (1)****Q39 (2)****Q40 (1)****Q41 (6)****Q42 (5)****Q43 (7)****Q44 (0)****Q45 (2)****Q46 (4)****Q47 (1)****Q48 (0)****Q49 (5)****Q50 (6)**